6.1200 *Mathematics for Computer Science* 31 October 2026
Problem Set 10
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Problem 1. Random Pixels

(a) Since there are 4 colors, the total number of way to color 4 pixels is 4^4. Also, there are 4 ways to color all 4 pixels with the same color. Thus, the probability that all 4 pixels have the same color is:

$$Pr[A] = \frac{4}{4^4} = \frac{1}{4^3}$$

(Note: By symmetry, $Pr[A] = Pr[B] = Pr[C] = Pr[D]$)

(b) First let us calculate the $Pr[A \cap B]$ since a and b share 1 square there are total 4^7 ways to color the 7 pixels. Also, there are 4 ways total way to make all 7 pixles the same color. Thus, we have:

$$Pr[A \cap B] = \frac{4}{4^7} = \frac{1}{4^6}$$

Also, again by symmetry, $Pr[A \cap B] = Pr[A \cap C] = Pr[A \cap D] = Pr[B \cap C] = Pr[B \cap D] = Pr[C \cap D]$. Now, we can calculate $Pr[A \cap B \cap C]$ since a, b, and c share 2 squares; there are a total of 4^{10} ways to color the 10 pixels. Also, there are 4 total ways to make all 10 pixels the same color. Thus, we have:

$$Pr[A \cap B \cap C] = \frac{4}{4^{10}} = \frac{1}{4^9}$$

Now Proof that A,B and C are 3-wise independent. We have:

$$Pr[A \cap B] = \frac{1}{4^6} = \frac{1}{4^3} \cdot \frac{1}{4^3} = Pr[A] \cdot Pr[B]$$

Similarly, we optain $Pr[A \cap C] = Pr[A] \cdot Pr[C]$ and $Pr[B \cap C] = Pr[B] \cdot Pr[C]$. Also, we have:

$$Pr[A \cap B \cap C] = \frac{1}{4^9} = \frac{1}{4^3} \cdot \frac{1}{4^3} \cdot \frac{1}{4^3} = Pr[A] \cdot Pr[B] \cdot Pr[C]$$

Thus, we have shown that A, B, and C are 3-wise independent.

(c) Now we can calculate $Pr[A \cap B \cap C \cap D]$ since a, b, c, and d share 4 squares, there are total 4^{12} ways to color the 12 pixels. Also, there are 4 total ways to make all 12 pixels the same color. Thus, we have:

$$Pr[A \cap B \cap C \cap D] = \frac{4}{4^{12}} = \frac{1}{4^{11}}$$

Now we can see that:

$$Pr[A \cap B \cap C \cap D] = \frac{1}{4^{11}} \neq \frac{1}{4^3} \cdot \frac{1}{4^3} \cdot \frac{1}{4^3} \cdot \frac{1}{4^3} = Pr[A] \cdot Pr[B] \cdot Pr[C] \cdot Pr[D]$$

Thus, we have proven that A, B, C, and D are not mutually independent.

Problem 2. Median vs Maximum

(a) There are a total of $\binom{100}{3}$ ways to select 3 numbers from 100. Now let us consider the case where k is the median. In this case, we need to select two numbers, one of which is less than k, and the other is greater. Thus, there are $\binom{k-1}{1} \cdot \binom{100-k}{1}$ ways to do this.

$$\text{PMF}_D(k) = \frac{\binom{k-1}{1} \cdot \binom{100-k}{1}}{\binom{100}{3}} = \frac{(k-1) * (100-k)}{\binom{100}{3}}$$

Similarly, we can calculate the case where k is maximum. In this case, we need to select two numbers such that both of them are less than k. Thus, there are $\binom{k-1}{2}$ ways to do this.

$$\text{PMF}_X(k) = \frac{\binom{k-1}{2}}{\binom{100}{3}}$$

(b) There are a total of $\binom{100}{3}$ ways to select 3 numbers from 100. Now there are two ways to choose 3 3 mumbers such that the median is $\leq k$. Case 1: All 3 numbers are $\leq k$; there are $\binom{k}{3}$ ways to do this. Case 2: Two numbers are $\leq k$ and last one is $> k$, there are $\binom{k}{2} \cdot \binom{100-k}{1}$ ways to do this. Thus, we have:

$$\text{CDF}_D(k) = \frac{\binom{k}{2} \cdot (100-k) + \binom{k}{3}}{\binom{100}{3}}$$

(c) The answer is 0 because the median cannot be 10 while the maximum is 5.

(d) No, they are not independent because $Pr[D \cap X] = 0$, meaning they are mutually exclusive.

(e) Value of PDF will be > 0 for $k \in \{1, \ldots 98\}$
because maximum difference between X and D is 100 - 2 = 98. and minumum difference between X and D is 100 - 99= 1.

Problem 3. Probabilistic Method

(a) $Pr[C]$ = Probability of at least 1 literal is true = 1 - none of them are true = $1 - \frac{1}{2^k} = \frac{2^k-1}{2^k}$.

(b)

$$U(\phi) = \sum_{i=1}^{n} U_i$$

Where

$$U_i = \begin{cases} 0, & \text{if } C_i = \text{True} \\ 1, & \text{if } C_i = \text{False} \end{cases}$$

(c) $Pr[U_i = 1] = P(C_i = \text{False}) = \frac{1}{2^k} = \frac{1}{2^k}$

Now, by linearity of expectations

$$Ex[U] = \sum_{1}^{m} Ex[U_i]$$

Also since U_i is indicator variable $Ex[U_i] = Pr[U_i]$.

$$Ex[U] = \sum_{1}^{m} Pr[U_i = 1] = \frac{m}{2^k}$$

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(d)

Proof by contradiction. Suppose $Pr[U = 0] = 0$, which means there will always be at least 1 clause that is unsatisfied; therefore $Ex[U] \geq 1$. But by using the result of part(c) and plugging $k = 5$ and $m = 31$, we get $Ex[U] < 1$, which is a contradiction. □

(e) From part (d), we established that $Pr[U = 0] > 0$. This implies that there exists some assignment of variables such that every clause is satisfied when $k = 5$ and $m = 31$.

Problem 4. Teeter Dance

(a)

	location								
	-4	-3	-2	-1	0	1	2	3	4
initially					1				
after 1 step				1/2	0	1/2			
after 2 steps			1/4	0	2/4	0	1/4		
after 3 steps		1/8	0	3/8	0	3/8	0	1/8	
after 4 steps	1/16	0	4/16	0	6/16	0	4/16	0	1/16

(b)

(i) $s = L_t = i - (t - i) = 2i - t$, Because if Spinda walked i steps to East then he must have walked $t - i$ steps to West.

(ii) There are a total of i east steps and $t - i$ west steps, so the total number of ways to arrange east and west steps is $\binom{t}{i}$ because out of t steps we just need to choose i positions where Spinda goes east, and on the remaining steps he goes west.

(iii) Since we are taking t steps probability of and probability each step is $1/2$ with general product rule probability that Spinda follows a particular path of length t is $1/2^t$, and from part (ii) we know there are a total of $\binom{t}{i}$ such paths; therefore probability that Spinda ends at this same location s after t steps is:

$$\frac{1}{2^t} \cdot \binom{t}{i}$$

(iv) From part (i) we know $L_t = 2i - t$ So, $B_t ::= (L_t + t)/2 = 2i - t + t/2 = i$.

B_t describes the number of steps taken in the east direction out of a total of t steps. Since each step taken by Spinda is independent and the probability of each step is $1/2$, and we are taking a total of t steps, it follows a binomial distribution $\text{Bin}_t(1/2)$.